



Summer 2024 ROUND 1 August 2-5, 2024

Instructions and Information

1. Teams can consist of up to 4 people. Each team should sign up before Round 1 through the procedure stated in the primer.
2. This is a 20-question, 3-day test. All answers are integers. Each question has a given weight (number of points) shown in the test beside the problem. The maximum number of points you can get is 50.
3. Please input your answers through the `/answer_question` command in the DSMC Contest Bot. Submit your answers through the `/submit_all_answers` command; by using this command, you may not edit your answers.
4. No calculators, books, graphing software, online resources, etc. are allowed. However, we grant permission to use word processing applications (collaborative or not), e.g. Google Docs, Overleaf, etc. so you may collaborate with your team. For physical supplies, only blank scratch paper, ruler, compass, and writing materials are allowed. If anyone is caught with illegal resources, they will be disqualified.
5. All communications must be done in the team channels in the Mathematics server.
6. No person who participated or helped in the contest is allowed to discuss any of the problems before the round ends. If moderators catch anyone discussing during this time frame, they will automatically be disqualified.
7. The cutoff score for Round 2 is the median of all scores.

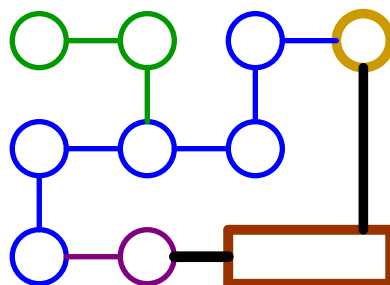
§1 Problems

- [1] Find the value of the digit x in which the number $\overline{345x}$ is divisible by 12.
- [1] If p, q are relatively prime integers with $q > 0$ such that

$$\frac{p}{q} = \sum_{i=0}^{\infty} \frac{3^i - 2^i}{5^i}$$

Find $p + q$.

- [1] A number p in base 10 can be written as $\overline{mn}$ in base b , where b is a positive integer at least 2 and n is not 0. If $6p$ can be written as $\overline{n04}$ in base 7, x is the sum of all possible values of p , and y is the sum of all values of m , find the value of xy .
- [1] $ABCD$ is an isosceles trapezoid where $AD \parallel BC$, $AB = CD = 13$, $AC = BD = 15$, and $AD = 4$. Find $[ABCD]$.
- [1] In the circuit below, the yellow circle represents a lightbulb, the rectangle represents a battery, the other circles represent switches, and the line segments represent connections between the objects. Each of the eight switches can only turn two ways: the purple switch can turn left or upward, the blue switches can turn right or upward, and the green switches can turn right or downward. Any two adjacent switches cannot point to each other. The lightbulb will only light up if there is a cycle that is formed with the connections that go through the lightbulb, the battery, and some switches. An example where the lightbulb lights up is shown in the diagram. How many configurations of the switches cause the lightbulb to light up?



- [1] $ABCD$ is a trapezoid with $AB \parallel CD$. AC intersects BD at X . A line passing through X and parallel to the two parallel sides of $ABCD$ is drawn and meets AD and BC at E and F respectively. If $AB = 7$, $CD = \frac{p}{q}$, $EF = 8$ for relatively prime positive integers p, q , find $p + q$.
- [2] Let N be a positive integer satisfying the following properties:
 - N is a perfect cube.
 - The last six digits of N are not all zeroes.
 - If the last six digits of N are removed, the number formed is also a perfect cube.

Determine the maximum possible value of N .

8. [2] Kaiser is giving a multiple-choice type exam. He can attempt a problem in three ways: fully confidently, half confidently or without knowing it at all. The probability that the answer is correct given he has answered fully confidently is $\frac{9}{10}$. The probability that the answer is correct given he has answered half confidently is $\frac{2}{3}$. The probability that the answer is correct given he has answered without knowing it is $\frac{1}{5}$.
- The probability that Kaiser doesn't know a question's answer is $\frac{3}{20}$. Given that he attempts all the questions, and the probability of a question being answered without knowing by Kaiser given that it is correct is $\frac{9}{221}$. If the probability that Kaiser answers a question fully confidently is $\frac{a}{b}$ for relatively prime positive integers a, b , find $100a + b$.
9. [2] Let $\{a_n\}_{n \geq 0}$ be a sequence with $a_0 = 27$ and $a_{n+1} = 37a_n^2 + 2a_n$ for nonnegative integers n . Evaluate $\lfloor \log a_{20} \rfloor$.
10. [2] $\triangle ABC$ has circumradius equal to $3\sqrt{3}$. X, Y, Z are points on sides BC, AC, AB respectively such that AX is median and BY is angular bisector of $\angle ABC$, and AX, BY, CZ are concurrent line segments. The lengths of AB, AC are $3\sqrt{3}, 9$ respectively. The inradius of $\triangle AZC$ is r . If $\frac{27}{r}$ can be written as $a + b\sqrt{c} + d\sqrt{f}$ where a, b, c, d, f are integers and $c \neq f$. If c, f are square free then value of $a + b + c + d + f$.
11. [2] Let sets $A = \{x : x = 13n + 7; n \in \mathbb{Z}; n \in [0, 305]\}$, $B = \{x : x = 5m - 1; m \in \mathbb{Z}; m \in [24, 375]\}$. If there are N possible functions $f : A \rightarrow B$ is a function such that:
- f is a non-decreasing function;
 - For integers λ such that $6\lambda \in A$, $f(6\lambda)$ is divisible by 7;
 - For unequal integers λ_1, λ_2 such that $6\lambda_1, 6\lambda_2 \in A$, $f(6\lambda_1) \neq f(6\lambda_2)$.
- Find the number of positive divisors of N .
12. [2] The sum of digits of n is 2024. Find the sum of the possible values of the sum of digits of $2n$.
13. [3] $\triangle ABC$ has $AB = 10, AC = 12$. X is on BC such that AX bisects $\angle BAC$. D is a point on ray AX such that $CD = 18$. If the maximum value of $\angle CDA - \angle BDA$ is θ , and $\sin^2 \theta = \frac{p}{q}$ for relatively prime positive integers p, q , find $p + q$.
14. [3] Culver is walking around the plane. He starts from the origin and makes 10 random moves, each 1 unit long, in one of the 4 principal directions. Given that he returns to the origin in the end, the probability that he returns to the origin exactly twice can be written as $\frac{p}{q}$ for relatively prime positive integers p, q . Find $p + q$.
15. [4] x, y, z are rational numbers such that

$$\sqrt[3]{\sqrt[3]{2} - 1} = \sqrt[3]{x} + \sqrt[3]{y} + \sqrt[3]{z}$$

If $xyz = \frac{p}{q}$ where p, q are relatively prime integers with $q > 0$, find $p + q$.

16. [4] For any positive integer n , let $f(n)$ denote the maximum positive integer m such that there exists real constants $a_{i,j}$ where $1 \leq i \leq n$ and $1 \leq j \leq m$ with absolute value 1 such that for any reals $x_1, x_2, \dots, x_m$, the following holds true:

$$\sum_{i=1}^n \left(\sum_{j=1}^m a_{i,j} x_j \right)^2 = n \sum_{j=1}^m x_j^2$$

Find

$$\sum_{k=0}^{10} f(2^k)$$

17. [4] $\triangle ABC$ has centroid G and incenter I . D is a point of side BC . Lines AD, IG intersect at M . BM intersects side AC at E . If $BD = 4, CD = 6, CE = 3, AB - AC = 2, AE = \frac{p}{q}$ where p, q are relatively prime positive integers, find $10p + q$.

18. [4] A triple of integers (a, b, c) is called *nice* if

$$\frac{5}{a} + \frac{5}{b} + \frac{5}{c} - \frac{2a}{bc} - \frac{2b}{ca} - \frac{2c}{ab} = \frac{27}{2(a+b+c)} - \frac{2025}{abc(a+b+c)}.$$

Let C be the number of nice triples, M, m be the maximum and minimum of $a+b+c$ over all nice triples. Find $1000000C + 1000M + m$.

19. [5] A certain complex number α satisfies the cubic equation

$$x^3 + bx^2 + cx + d = 0,$$

where b, c, d are integers, whereas α^4 satisfies the cubic equation

$$x^3 - \gamma x - 16 = 0$$

where γ is an integer. Both equations do not have integer solutions. Given that there are exactly 6 possible values for γ , find the sum of these values.

20. [5] Culver and Kaiser play a game. There are 2024 stacks of stones, arranged from left to right, such that the number of stones in each stack is a positive divisor of 2024^{1024} . Each turn, with Culver starting first, the person must choose one of the stacks with more than one stone, and split it into multiple (more than one) stacks with equal number of stones. The player who cannot move loses. Compute the number of initial configurations such that Kaiser has a winning strategy, modulo 10000.

§2 Answers

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|--------------|------------|---------------|
| 1. 6 | 8. 305 | 15. 721 |
| 2. 11 | 9. 3145726 | 16. 2047 |
| 3. 68 | 10. 28 | 17. 41 |
| 4. 108 | 11. 777801 | 18. 108149922 |
| 5. 37 | 12. 903150 | 19. -646688 |
| 6. 31 | 13. 82 | 20. 8625 |
| 7. 125751501 | 14. 5347 | |